COLDSTART: Using published evidence to forecast health-system crises when local outcome data are still too scarce

1. Summary

When a health crisis begins, the quantities decision-makers need to forecast (calls, presentations, demand, occupancy, severity) are precisely those for which almost no local outcome data yet exist. The outcome series is a handful of noisy observations, and systems that perform well on years of local history are weak when early decisions matter most. Published evidence from analogous events elsewhere is another source of quantitative information. It is rarely tested as formal prior information, because of the lack of reliable extraction and the difficulty of transporting it to the new setting. A confidently wrong prior is worse than none, and it is most influential exactly when local data cannot correct it.

This project asks one question. **At the start of a crisis, when local outcome data are still too scarce, can published quantitative evidence, extracted automatically, adapted to the local setting and fed into predictive models, improve forecasts of health-system escalation?** Answering it means showing that the evidence is extracted reliably, that we can tell early when it misleads, and that any gain is large enough to change decisions. Finally, the whole system runs on its own, with models built from the evidence and its forecasts verified prospectively against observed outcomes. **LiteRev-Evidence**, the prototype I built, already covers this whole path. Starting from a question in plain language, it extracts the published numbers and grades how certain they are, combines them into distributions of plausible values, suggests explanatory variables, and then fits and compares candidate models.

The project turns this prototype into a validated instrument and tests it in four steps. First, I measure the errors of automated extraction and carry them into the priors, for parameter values and everything else the literature reports (explanatory variables and their delays, model types, outcomes and alert thresholds). Second, I describe the care system as a hidden level of strain (routine, elevated, strained or critical) whose probabilities carry the uncertainty of every input. The model behind it is fed by the evidence and by an early-warning signal from the short local history, the critical slowing down that often precedes a tipping point. Third, I replay past crises day by day, using only the information available at each moment, to test whether the evidence improves early forecasts and when it does harm. Fourth, I test whether any gain is large enough to change operational decisions. In the decisive experiment, an automated engine checks and scores a library of candidate models registered in advance (local baselines, surveillance algorithms, machine-learning and regime models, and transmission models for epidemics only). It picks the best model in each of two arms, with and without the evidence, by rules fixed in advance. The confirmatory test compares the two on respiratory epidemics and, only if the evidence wins there, on heatwaves.

The forecasting tools are validated first on open Swiss data, released as a public benchmark, then on Geneva's operational data. Three types of crisis in Geneva, and no more, carry the real-world tests. Respiratory epidemics carry the main test, heatwaves the second, and legionellosis a limited extension in year 4. Both results are useful. If the evidence helps, the project shows when and by how much. If not, or if harm cannot be caught early enough, it sets a clear limit for automated evidence-informed forecasting. Where models pass validation and use is authorised, the system runs by month 48 as a **daily dashboard** with Geneva's emergency services for demand-driven crises, observing only and never steering care. It is the core of a future living surveillance platform.

The project rests on an unusual combination of fifteen years of quantitative finance on rare-event risk and regime models, LiteRev-Evidence, and Swiss emergency data. Ambizione turns these into an independent research programme that I lead.

2. Proposed research

2.1 Current state of research in the field

2.1.1 Forecasting health-system crises is strong in steady state and weak at onset

Syndromic surveillance detects departures from expected baselines [Farrington 1996; Noufaily 2013], and forecast ensembles perform strongly when data are plentiful [Cramer 2022; Sherratt 2023]. Dispatch records move before confirmed surveillance. The 2009 A(H1N1) autumn onset was identified eight days early in three European regions [Rosenkötter 2013], long call series track influenza-like illness [Bonora 2025], and our systematic review maps the field [Edjinedja 2026].

The decisive limitation is **dependence on history**. Data-driven models need enough local observations to learn seasonality, the response to weather and the dynamics of a crisis. Reviews of emergency-demand forecasting list which model families work for which outcome and horizon [Wargon 2009], and every recommended model is trained on years of local history.

The cold-start problem is a missing-outcome problem. At the start of a crisis, context variables abound but the outcome to be forecast does not, and abundance itself can mislead. Google Flu Trends overestimated influenza by more than a factor of two [Lazer 2014]. Early signals do not close the gap. Wastewater needs a conversion from viral shedding to incidence that itself comes from external evidence, and transfer learning needs comparable observations from the same period. The common issue is **how to use external quantitative information without pretending it transfers perfectly**.

2.1.2 Published evidence is an unused source of prior information

Bayesian borrowing is well developed, with power priors [Ibrahim 2000], commensurate priors [Hobbs 2011] and meta-analytic-predictive priors that protect against conflict [Schmidli 2014]. Informative priors have been used to forecast disease when local data are sparse [Cook 2023]. What has not been established is narrower and harder. Does an evidence prior that is **built automatically**, and carries both its extraction error and the uncertainty of transferring it, improve **operational forecasts at cold start when only past information is allowed**? And is harmful borrowing detected in time to act?

Parameter values are not the only thing that transfers. The literature also reports **explanatory variables and their delays**, **evidence on model forms** (which algorithm families work for which crisis type and horizon) and **definitions of outcomes and thresholds**. Living systematic reviews keep syntheses up to date [Elliott 2014], yet to my knowledge no operational forecasting framework uses these elements systematically.

Extraction. Automated extraction makes large-scale synthesis feasible, but extracting numbers remains less reliable than extracting categories (reported accuracy roughly 47–88% versus 74–96%), and omissions are the main source of error [Shankar 2026]. The open question is whether extraction and pooling keep the spread of values that a calibrated prior needs.

Transportability. Even perfectly extracted estimates may not transfer across populations, case definitions, health systems or policies. Formal tools for transportability exist [Bareinboim 2016; Dahabreh 2020; Degtiar 2023], but no operational cold-start framework uses them.

2.1.3 Escalation can be represented as a state, not only as a point forecast

Emergency services often need a clear level of escalation (routine, elevated, strained or critical) more than tomorrow's count. Hidden-state models are established in surveillance. Poisson hidden Markov

models separate epidemic from non-epidemic periods [Le Strat 1999; Watkins 2009], and extreme-value methods describe rare critical peaks [Coles 2001], including hospital congestion in Switzerland [Ranjbar 2022]. Here they serve one purpose, a hidden ordered state that gives a **common frame for testing evidence borrowing at the start of a crisis**. Mechanistic transmission models [Keeling 2008] enter only as components when the crisis is an epidemic, because environmental crises such as heat are driven by exposure and response, not by transmission. Critical slowing down describes how variance and lag-1 autocorrelation rise before some transitions [Scheffer 2009], with applications to epidemics [O'Regan 2013; Brett 2018; Southall 2021]. It reads the shape of a short recent series and is tested empirically as a **secondary source of information**.

2.1.4 Forecast accuracy is not yet decision value

Proper scoring rules reward calibration [Gneiting 2007], but decision-analytic methods judge predictions by their explicit consequences [Vickers 2006]. A forecast is useful in practice only when its probabilistic gain changes a defensible decision. Heat thresholds calibrated on mortality need not match thresholds for demand [Lung 2021], and links between pollution and asthma emergencies differ across settings [Zheng 2015]. Methods are therefore judged first statistically, then under a loss structure elicited from the services.

2.1.5 The specific gap addressed by this project

Ongoing programmes occupy the neighbouring ground. Multi-model forecasting hubs run routinely (ECDC RespiCast, US CDC FluSight), Horizon Europe funds infrastructure for epidemic intelligence (including GeoAI4EI, in which I take part), and the Franco-Swiss GESICA programme builds cross-border crisis intelligence. This project does not compete with them where data are plentiful. None of their stated objectives asks whether automatically synthesised external evidence should enter a forecast as formal prior information at local cold start. Nor has anyone shown that, around this **cold-start question**, the loop from living evidence synthesis to variables, models, thresholds and automatically re-validated forecasts can be closed under rules registered in advance.

2.2 Current state of personal research and competences required for the project

My route is unusual, and it is why this project is feasible. I spent fifteen years in quantitative finance (risk, extreme values, regime models, stress testing), then completed a doctorate in biomedical sciences at Geneva (defended 18 December 2023).

2.2.1 Prediction under sparse and imperfect information

In [Orel 2022] I predicted individual HIV status from socio-behavioural characteristics across East and Southern Africa and established where models transferred between countries. [Merzouki 2021] and [Esra 2023] developed the theme. I now **senior-author** that line, as last author of [Ng'ambi 2026] (accepted) on cardiovascular-disease classification across harmonised WHO STEPS surveys. I first met this proposal's transportability problem elsewhere, and my role on it has moved from conducting to directing.

2.2.2 Automated evidence synthesis

In [Orel 2023] I introduced **LiteRev**, an automated literature-review tool that searches open databases with a natural-language or Boolean query, removes duplicates, maps the corpus and suggests relevant papers step by step. I led its development with **Aziza Merzouki** (PhD, computer science) and raised its funding on my own initiative, with **CHF 30,000** (UNIGE), **CHF 10,000** (Venture Kick) and **CHF 20,000** (Mimosa), outside any group grant. The systematic review of AI in emergency medical services [Edjinedja 2026], within GESICA, uses it in its methods.

2.2.3 Outbreak and health-system modelling in Switzerland

In [Orel 2024] I compared clinical severity between Delta and Omicron sub-lineages in a Swiss tertiary centre. [Estill 2020] built age-structured scenario models for the Swiss SARS-CoV-2 epidemic under planning time pressure. That experience is where this proposal's question comes from. I contributed to WHO African Region reporting and seroprevalence estimation [Nwosu 2021].

2.2.4 The prototype and test bed, LiteRev-Evidence

Since 2024 I have developed **LiteRev-Evidence**, which extends LiteRev into structured quantitative extraction and modelling. The full review pipeline works end to end. It searches the open literature live, removes duplicates, screens with PRISMA accounting and extracts PICO elements. It then extracts numbers with their source, **grades them for the certainty of the study design and pools them into parameter distributions**. At the model step it proposes candidate explanatory variables with their supporting articles and typed outcome definitions with ready-made emergency-demand targets (ED overload, bed occupancy, call surge). It **fits models across algorithm families with automated hyperparameter search, validates them on time series without leakage and ranks them on a stated metric**. It also builds compartmental components from extracted ranges, usable as predictors. **Living reviews re-run on schedule** for each scenario, and every update is previewed before it is applied. This is the evidence-to-model loop this proposal examines, in prototype form.

For GESICA I built the data foundation of the Geneva–Vaud–Neuchâtel network, and COLDSTART uses its Geneva core. It covers **77 notifiable diseases in eight model classes** and a referenced inventory of **28 surveillance sources (23 infectious, 5 environmental or non-infectious)**, each documented for institution, coverage, resolution, delay, access and quality. The data landscape is therefore mapped, not assumed, and validation domains are chosen by **model class**.

The prototype makes the research feasible rather than aspirational. The fitting, tuning and ranking that T3.3's selection engine registers already run in early form. The grant adds the science (two arms, registered rules, inference at the level of episodes), not the plumbing, and the validated instrument is itself a deliverable.

2.2.5 Linked data on a contrasting type of crisis

I lead the data work on a cantonal study **already under way** (BASEC 2026-00324, ethics granted) that links confirmed legionellosis cases in Geneva to individual domestic hot-water installations. This distinctive linkage supplies a contrasting waterborne type of crisis.

Competences. Bayesian hierarchical and regime-switching estimation, extreme-value modelling, stress testing under misspecification, forecast evaluation and decision analysis come from my work in risk and form the core of WP2 and WP3. Extraction and pooling come from LiteRev-Evidence. Domain knowledge comes from the doctorate, the Swiss COVID-19 work and GESICA.

2.3 Detailed research plan

2.3.1 Objectives and hypotheses

The aim is to determine **whether, and under what conditions, published evidence can support forecasting at the start of a health-system crisis**. This question is **H3a, the central hypothesis**, and every other row tests one link it depends on.

	Role	Statement
H1	Validation	Can the evidence be trusted enough to use?
C2	Adequacy criterion	Is the state model fit to compare borrowing in?
H3a	**Central hypothesis**	**Does systematically extracted evidence improve cold-**

	Role	Statement
		start forecast skill?
H3b	Robustness	Is adaptive borrowing safe when the prior is wrong?
H3c	Secondary channel	Do resilience indicators add information beyond the evidence?
H4	Decision value	Is the gain large enough to change an operational choice?

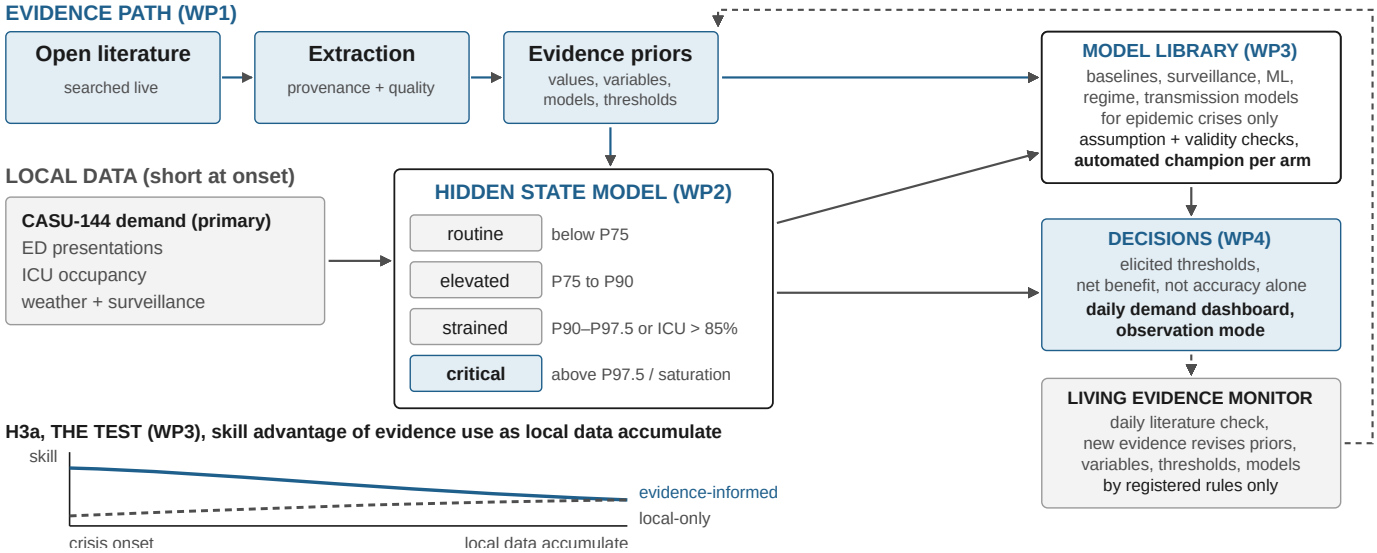


Figure 1. The cold-start problem, the evidence pipeline, the model library with automated selection, and the living update loop.

O1. Make published evidence usable without hiding its uncertainty

H1. Automated extraction will systematically **understate the spread of the evidence-derived distribution** (reported within-study uncertainty and between-study heterogeneity) and so produce priors that are too narrow. An explicit measurement-error layer will correct this under-dispersion. Success is judged by a criterion set in advance, the coverage and calibration of intervals from the corrected distribution across held-out adjudicated studies.

Omissions are the main reported extraction error [Shankar 2026], and H1 tests exactly whether they systematically shrink the spread. If the loss cannot be corrected, the project establishes a limit. The evidence covers four classes of objects (T1.1). H1 is tested on the quantitative core, and classes 2 to 4 are validated externally against published syntheses (T1.2).

O2. Represent escalation as a state, separate from the point forecast

C2, model adequacy criterion. The hidden-state model must give **identifiable** parameters and **calibrated** probabilities of each escalation state at matched false-alarm rates. Its role is to provide the common frame in which borrowing strategies are compared. It does not claim that regime switching generally beats a threshold on a point forecast.

T2.1's identifiability study and T3.3's calibration checks either establish adequacy or trigger the ordinal fallback, and either outcome leaves H3a intact. Extreme-value modelling covers the critical tail, and resilience indicators are **supporting covariates**.

O3. Test the cold-start hypothesis and map failure

H3a. A forecasting configuration informed by the evidence (evidence-derived priors, sets of explanatory variables and model forms) improves probabilistic forecast skill in the early phase of a crisis. How long the advantage lasts, and how it fades as local observations accumulate, is assessed as a secondary question.

H3b. Adaptive borrowing, which discounts the evidence when the prior and the data conflict, is **non-inferior** to fixed borrowing when the priors are well specified, within a margin Δ set in advance on the CRPS skill score (the CRPS is a standard score for probabilistic forecasts). It is **superior** to fixed borrowing when the priors are deliberately misspecified.

Δ is fixed at half of the minimal relevant H3a improvement, a 50% preservation fraction set in advance. Adaptive borrowing must keep at least half the gain that justifies borrowing at all. Δ is therefore a maximum acceptable loss, not a convenience of the simulation.

H3c. Resilience indicators add predictive information beyond the evidence-derived prior and the local level and trend when the outcome history is short.

Primary confirmatory comparison, stated once. The **best evidence-informed model (the champion)** is compared with the **best local-only model**, both chosen by the same automated procedure registered in advance (T3.3). The measure is the **CRPS skill score** over the cold-start window set in advance, **on respiratory episodes only**. Heat repeats the same comparison as a **sequential generalisation test**, run only if the respiratory test is passed. This fixed order controls the family-wise error rate. The report also shows **how the advantage changes as local data accumulate**, and it should fade to nothing.

O4. Establish whether better prediction matters for decisions

H4. For the demand-driven crisis types, a decision-analytic evaluation that uses the losses and escalation thresholds of emergency responders can rank modelling strategies differently from generic accuracy measures. The evidence-derived strategy is useful only when its gain crosses a decision threshold.

Scope. Three types of crisis in Geneva, fixed for the grant

COLDSTART's real-world scope is three types of health-system crisis, and no more. (1) **Respiratory epidemics** carry the primary confirmatory test. SARS-CoV-2, influenza and RSV overlap in their symptoms and are monitored together in Switzerland. (2) **Heat events** carry the sequential generalisation test, with **air pollution (ozone, PM10) as a co-exposure and effect modifier**, not a separate domain. (3) **Waterborne outbreaks**, through Geneva legionellosis, form the contrasting extension in year 4 (T3.5). Further data serve only benchmarking and sensitivity analyses. For respiratory epidemics and heat, both arms score **the same quantity, built the same way**. It is daily emergency demand from the CASU-144 series, restricted to the cause classes of each crisis type (T3.0). For epidemics this is respiratory-related demand. For heat it is **heat-sensitive demand**, since Swiss evidence places heat effects in dehydration, renal and psychiatric admissions, with a weak respiratory effect at older ages [Schulte 2024; Ragettli 2019]. Legionellosis instead forecasts case incidence (T3.5), outside this comparison of demand. If the month-12 gate activates the registered fallback, both arms score that fallback, built the same way, and the claim narrows.

2.3.2 Work packages and methods

WP1. From published evidence to usable priors (M1–M20)

T1.1. Define the evidence target and register the protocol *(M1–M4)*

The evidence covers four classes of objects. Inclusion criteria, effect measures, the representation of uncertainty and the transportability variables are fixed before extraction begins. **(1) Parameter values form the quantitative core and are benchmarked in full.** They include **weather-demand coefficients with their delays** (daily mean temperature, heat-day exceedance, ozone, PM10), **the size and timing of surges** (peak-to-baseline ratio, time to peak, growth rate at onset), **distributions**

of length of stay and occupancy (ward and ICU) and **admission fractions**. Transmission parameters (R0/Rt, serial interval) and conversions from viral shedding to incidence are secondary. **(2) Explanatory variables and their delays** reported as informative for each crisis type. **(3) Evidence on model forms**, that is, which algorithm families work for which outcome, horizon and crisis type [Wargon 2009], feeding T3.3. **(4) Definitions of outcomes and thresholds**, that is, which series services use and at what levels warnings fire [Lung 2021], as starting points for T2.1's anchors and T4.1's elicitation. Classes 2 to 4 carry their source and certainty grades and are pursued only as inputs to the registered H3a configuration. H1's benchmark applies to class 1.

T1.2. Build the quantitative extraction benchmark *(M3–M9)*

A separately contracted independent extractor (§2.4) and I each extract the target quantities from a stratified random sample of **300** publications, and differences are adjudicated. The strata cover parameter class, reporting quality, study design and **source type** (journal versus timestamped situation report), so the error is measured separately for T3.2's faster secondary sources. An **open external layer** scores retrieval against the inclusion lists of published systematic reviews in the target classes, and extraction and pooling against their published summary estimates. The benchmark is an open deliverable.

T1.3. Characterise automated extraction error *(M7–M14)*

Automated extraction is compared with the benchmark on point estimates, uncertainty, omissions and between-study spread. H1's primary measure is the median log-ratio of the automated to the adjudicated **standard deviation of the evidence-derived distribution**. It is computed per parameter class on a common scale under the registered pooling model, with an interval set in advance, and a ratio below one means under-dispersion. Within-study uncertainty and between-study heterogeneity are carried through separately and together, with sensitivity to the model version.

T1.4. Construct uncertainty-aware evidence distributions *(M12–M18)*

Extraction error is represented explicitly as measurement error. It compares inverse-variance and quality-weighted pooling, meta-analytic-predictive priors and power-prior discounting. The output is a prior with its source, its extraction uncertainty and a documented borrowing weight.

T1.5. Screen for transportability *(M15–M20)*

It describes the effect modifiers and how study settings differ from Geneva. The transportability criteria, and the rule that turns a transportability score into a wider or discounted prior, are fixed before the confirmatory evaluation. This produces the metadata WP3's conflict analysis needs.

Deliverables. D1.1 open benchmark, D1.2 error analysis, D1.3 evidence-to-prior library.

Risk. If expert time runs short, the benchmark narrows to the core parameter classes.

WP2. A simple model of health-system escalation (M1–M28)

T2.1. Specify the state model and check it can be identified *(M1–M9)*

A Bayesian hierarchical Markov regime-switching model [Hamilton 1989] has a hidden ordered state (routine, elevated, strained, critical) observed through emergency calls, the primary channel, with ED presentations and ICU occupancy added where available. Weather, calendar and epidemic covariates are set in advance, and each series has its own observation model linked to the shared state. Before fitting real data, simulations test whether the parameters and states can be recovered, including when fewer channels are available. Ordering constraints resolve label switching. If the states separate poorly, the fallback is an ordinal state-space model. The criterion is the recovery of states and transitions, not visual fit. **Provisional state anchors** are fixed at registration for WP2 and WP3. T4.1's

later elicitation refines what they mean for decisions, never the confirmatory definition of the states. The state is **routine** below the 75th percentile of the seasonal baseline and **elevated** from the 75th to the 90th. It is **strained** from the 90th to the 97.5th, or under sustained capacity pressure (ICU occupancy above 85%, or ED boarding above its seasonal 90th percentile). It is **critical** above the 97.5th, or at capacity saturation. The highest applicable state prevails, and capacity saturation alone defines critical. Anchors are percentiles for each event type, and in estimation they enter as priors on the level of each state, not as hard cutoffs.

T2.2. Represent the critical tail *(M6–M14)*

Rare exceedances are modelled with peaks over threshold and the generalised Pareto distribution, linked to the probability of the critical state, with threshold sensitivity reported, in support of the other tasks rather than as a separate objective.

T2.3. Introduce evidence-derived priors *(M10–M20)*

WP1's distributions are mapped to the parameters where published evidence is relevant (weather effects, surge sizes, features of transitions). It compares weakly informative priors, fixed evidence-derived priors and adaptive robust borrowing. The adaptive version is a robust mixture of an evidence-derived and a weakly informative component. The conflict statistic and the discount function are fixed before the confirmatory evaluation and are not tuned during it. Power priors and commensurate priors serve as comparators in sensitivity analyses.

T2.4. Add resilience indicators as a secondary source of information *(M12–M20)*

Rolling variance and lag-1 autocorrelation enter as optional covariates on the transitions between states. The question is whether they add information beyond level, trend and the evidence prior. A null result is interpretable.

T2.5. Calibration, automation and validity monitoring *(M20–M28)*

It uses calibration methods suited to time-dependent data, with a conformal component [Angelopoulos 2023; Barber 2023] as a robustness layer if simulation confirms its assumptions. It also builds the **automation layer**. Public data streams (weather, surveillance, wastewater) are loaded automatically as they become available, and models are re-estimated at the natural pace of each input, with every re-fit versioned. Every fit runs **checks of its assumptions and validity** (residual structure, dispersion, calibration), and registered rules exclude a model that fails them. A reference implementation is released in LiteRev-Evidence.

Simulation is prior information, not data. Fitting trajectories built from literature parameters as if they were data would count the same information twice and disable the conflict check that H3b depends on. Simulation therefore serves identifiability, structural constraints and prior predictive checks, and never narrows the evidence prior.

Deliverables. D2.1 model specification and identifiability study, D2.2 open implementation, D2.3 methodological paper.

Risk. The states may separate weakly. *Mitigation.* Simulation comes first, the ordinal fallback is ready, and the identifiability limit is reported as a result rather than tuned until it succeeds.

WP3. The decisive cold-start experiment (M12–M42)

T3.0. Outcome hierarchy, data-access gate and episode eligibility *(M1–M14)*

All of the following is fixed before any evaluation is designed and never revisited because of performance.

Outcome. The primary outcome is **daily respiratory-related emergency demand derived from CASU-144 records**, not a raw count of calls. It rests on a registered classification of the recorded call reasons and urgency levels. **The heat arm scores heat-sensitive demand**, using the same series and construction restricted to cause classes fixed at registration (dehydration, renal, psychiatric), with sensitivity analyses restricted to respiratory causes and to people aged 75 and over. ED presentations and ICU occupancy enter as **additional observation channels on the shared hidden state**, and wastewater, sentinel consultations and weather enter as covariates.

Candidate outcomes and geography. After (1) the primary outcome, the registered candidates are, in order, (2) ED presentations by category, (3) ICU occupancy, and (4) all-cause 144 engagements and hospital admissions as sensitivity series. Open surveillance (Sentinella, wastewater) serves only as a fallback outcome. The unit is the **canton of Geneva, day by day**, which is the area the HUG dispatch centre covers. The single canton is deliberate, as I already work in its data ecosystem. Generalisation comes from the contrasting crisis types and the open national series.

Data-access gate. CASU-144 is the primary outcome set in advance. Criteria for **historical depth, resolution, delay and completeness** are also fixed in advance, and a gate at month 12 confirms them. If they are not met, the fallback set in advance activates and the claim narrows.

Episode eligibility. An episode enters the confirmatory evaluation only if all of the following can be rebuilt as they stood at the time. (1) An onset detectable under the real-time onset rule. (2) Enough history before onset for its rolling baseline. (3) Enough observations after onset at the primary horizon. (4) The external evidence **as it stood on the historical forecast date**. (5) No leakage of future information. (6) Separation from neighbouring episodes. Episodes that fail serve only sensitivity analyses. **At the level of demand, pathogens that circulate together form one episode.**

Two registration points. The order of outcomes, the gate criteria and the eligibility rule are registered now. The window N, the horizon for each crisis type and the margin Δ are registered after the checkpoint and the episode inventory, before any evaluation runs, using T3.3's selection rules declared in advance.

Open-data validation track (M12–M24, secondary, set in advance). Before any clinical series connects, the complete day-by-day replay runs on open Swiss series that need no agreement (federal respiratory surveillance, the national wastewater programme, weekly all-cause deaths, MeteoSwiss exposures). Archived forecasts from multi-model hubs give comparators from the same time for early rounds that were truly cold-start [Cramer 2022; Sherratt 2023]. The track is released as a **public benchmark that anyone can re-run**. Every component is thus validated before operational data arrive, whatever the access, and operational series then sharpen the claim to emergency demand.

T3.1. Assemble the retrospective information set *(M12–M20)*

The CASU-144 series is harmonised with the additional channels and covariates, and its completeness and delays are measured. Right truncation is modelled with nowcasting [Höhle 2014; McGough 2020], because reporting that degrades under strain could itself look like an early-warning signal. Two Geneva channels extend the set. The HUG COVID-19 hospitalisation data I analysed before [Orel 2024] are requested again as a secondary channel for respiratory validation and calibration. Pharmacy sales and wastewater measurements from the pharmacien cantonal serve as syndromic covariates.

T3.2. Rebuild the information available at each moment *(M18–M30)*

For each past onset, successive forecast dates (origins) use **only the information available on that date**. **A prior may only describe *other* populations, places or past events, whatever the venue. The confirmatory prior uses only peer-reviewed literature and preprints**, under a cut-off date

that moves with each origin. Timestamped situation reports on the same ongoing event *elsewhere* (WHO/ECDC) enter as a **secondary prior variant set in advance**. They pass through the same extraction pipeline but stay outside the confirmatory comparison, because H3a tests borrowing from the literature, not the fusion of all evidence. **Text about the local event is excluded from every prior**, because it is a noisy measurement of the outcome itself. Every input carries a timestamp. Revised sources enter as **the version released by the forecast date**, never the final revised value, with both observation and release times kept. The forecasts are re-run automatically through the T2.5 layer at each input's natural pace. Forecast horizons are 7, 14 and 28 days. The primary horizon for each crisis type is fixed at the second registration point by an operational criterion set in advance, never by forecast performance or episode results. The cold-start window counts the local outcome observations **after a real-time onset rule defined in advance** that uses only the variables available on the forecast date.

T3.3. The model library and the automated selection engine *(M20–M34)*

The library is registered before any evaluation and has two arms. The **local-only** arm holds a seasonal GLM with weekday and holiday terms, persistence, established surveillance methods for exceedance and nowcasting (Farrington and Noufaily, Bayesian nowcasting [Salmon 2016]), penalised, gradient-boosted and upper-tail quantile learners on the covariates set in advance, and the WP2 regime model with weakly informative priors. The **evidence-informed** arm holds the same candidates equipped with WP1's priors and design inputs, in three registered variants. These are fixed evidence-derived priors, adaptive borrowing with conflict monitoring, and adaptive borrowing plus resilience indicators (H3c). **Mechanistic transmission components [Keeling 2008] are allowed only for epidemics.** Environmental crises use exposure–response structures, and what is allowed for each crisis type is registered.

Automated choice of the champion, nested to prevent leakage. At each forecast date, a registered rule chooses each arm's **champion** using only an inner rolling validation on outcomes observed **before** that date. The CRPS is the main selection criterion. Log score and calibration are registered as conditions a model must meet, not as ranking criteria. Selection follows the T2.5 validity checks and is blind to the other arm. The chosen champion then issues the forecast that is scored. Selection never sees the outcomes it is scored on. Both arms use the same selection machinery, so the primary comparison tests the full evidence-informed configuration, not one model family. The library is broad on purpose, to give selection real choices, not to multiply hypotheses. Registered secondary comparisons isolate the contribution of the prior, the variable set and the model form.

Confirmatory testing procedure, in a fixed sequence.

1. **Test 1 (primary).** The evidence-informed champion against the local-only champion, on the CRPS skill score, for **respiratory** episodes over the cold-start window N, at $\alpha = 0.05$ two-sided, because harm from borrowing matters for decisions as much as benefit. Passing requires a paired-permutation $p < \alpha$ *and* a lower confidence bound above the registered minimal relevant improvement.
2. **Test 2 (generalisation).** The same comparison on **heat** episodes, scored on heat-sensitive demand (T3.0), at the same α, **run only if Test 1 passes**.
3. **If Test 1 fails**, H3a is not supported, Test 2 is exploratory, and the project's result is the failure map and the limit it sets.

All other comparisons are secondary. **The local-only arm is fixed in the registration.** Its regime-model priors come with a declared band of vaguer and tighter alternatives as a sensitivity analysis, so the comparator cannot be weakened after the fact.

Primary endpoint. Its aggregation is registered. For each episode and arm, the CRPS is averaged over the eligible forecast dates in the window at the primary horizon. The episode skill score is one minus the ratio of the two CRPS values (evidence-informed over local-only), and the confirmatory statistic is its equally weighted mean over episodes. The unit of inference is the **episode**. Eligible episodes are few (13–14 respiratory candidates before screening), so inference uses a **paired permutation over episodes**, with an **episode-level bootstrap** alongside that resamples whole episodes and so keeps their internal time dependence. Both are set in advance, and any disagreement is reported. **Simulation sizes the design at the level of episodes**, measuring type I error, power and interval width under realistic dependence before the second registration point. The same simulation fixes in advance the **minimum number of eligible episodes** below which no confirmatory claim is made and the primary analysis becomes estimation. It also fixes the **rule for choosing *N***, the smallest grid value that meets the operating criterion, chosen blind to forecast performance. The **minimal relevant improvement** is a registered value of the episode-level CRPS skill score, fixed before the second registration point by an operational-relevance criterion defined before any outcome is evaluated. T4.1's later elicitation translates observed improvements into operational consequences but does not set this threshold. **Δ is registered at half of this fixed value** (§2.3.1). The same simulation checks power for these targets and the stability of the ratio-based score when baseline CRPS varies between episodes. Secondary endpoints are the log score, calibration (PIT, coverage) and the detection of escalation at matched false-alarm rates. **H3b's Δ is fixed here, before any evaluation.** Non-inferiority means a CRPS deficit no greater than Δ, and superiority is tested on T3.4's misspecified priors. The well-specified case is defined in registered simulations where the distributions that generate the evidence and the data match within a set tolerance, and real episodes complement them.

T3.4. Map benefit and failure *(M28–M38)*

It identifies the episodes where borrowing improves or worsens forecasts. It describes failure by mismatch of population, outcome definition, health system, policy and time, by extraction uncertainty and by conflict between prior and data. Deliberately misspecified priors (registered shifts in location, scale and transport) serve as a stress test. The safety question is whether adaptive discounting catches harmful borrowing in time. The output is a **failure map**, not an average.

T3.5. Test generalisation *(M34–M42)*

This task runs the sequential heat test and, if resources allow, the legionellosis extension in year 4. The extension transfers the borrowing framework to forecasting case incidence at the start of an outbreak, with low counts and individual cases, on the series that links cases to their environment.

Deliverables. D3.1 reproducible cold-start evaluation pipeline, D3.2 primary result, D3.3 map of failures and stress tests, D3.4 analysis across crisis types.

Risk. Access to operational data. *Mitigation.* Agreements start before the award, and letters accompany the application. The open-data track already carries the methodological test under the same day-by-day restriction, so delayed access narrows the claim (§2.4) without stalling the project.

WP4. From predictive skill to operational value (M24–M48)

T4.1. Elicit operational losses and thresholds *(M24–M32)*

A structured **SHELF** elicitation with about 15–20 experts from HUG emergency medicine, CASU-144 dispatch and capacity management asks what early, late and unnecessary escalation cost, and derives thresholds from these losses. **The primary decision is fixed in advance.** Surge capacity is escalated when the forecast probability of the strained or critical state crosses the elicited threshold. Models without a state turn their forecasts into state probabilities through T2.1's anchors, so every model gives the same decision quantity.

T4.2. Decision-analytic evaluation and equity audit *(M30–M40)*

The WP3 forecasts are re-evaluated with net-benefit and decision-curve analysis and with value of information, to see whether rankings change once consequences count. Operational records may reflect structural differences, so calibration and threshold performance are checked across aggregate groups (age, sex, neighbourhood deprivation). A model that is calibrated on average but miscalibrated for a relevant group is not ready for operations.

T4.3. Counterfactuals, observation-mode dashboard and living updates *(M34–M48)*

For selected episodes, it estimates what would have changed if escalation had followed the model's signal. It uses a simple capacity model and carries the uncertainty through, so these are counterfactuals, not causal estimates. For each crisis type that passes T3.3's validation and T4.2's decision-value check, and subject to authorisation, the system runs as a **daily dashboard in observation mode** with the partner services (144 dispatch, ED, ICU). State probabilities, forecasts and validity checks refresh automatically and are **recorded and scored prospectively against the outcomes as they are observed, never used clinically**. The project ends with the system running and its forecasts verified. The legionellosis extension is methodological and retrospective, so it need not enter the capacity decision or the dashboard. A **living-evidence monitor** checks the literature daily and re-runs each scenario's syntheses on a registered schedule [Elliott 2014]. It flags new publications that would change a parameter value, a variable set, a threshold or the recommended model. Changes apply only through version-controlled update rules registered in advance, never silently. If authorisation is not granted, the project is complete on the retrospective evaluation and says so.

Deliverables. D4.1 elicited loss structure and equity audit, D4.2 decision-analytic evaluation, D4.3 counterfactual analysis, with the observation-mode dashboard and living-update protocol where authorised.

Methods, data protection and reproducibility. Code is version-controlled R and Python under a registered analysis plan, clinical data stay in the UNIGE secure environment under the required approvals, and the benchmark and software are open, with synthetic data where possible.

Expected outputs. Four to six papers and the durable open resources of §2.5. **I lead the methodological, benchmark and integrative outputs.** None of the fallbacks above turns an inconclusive analysis into a claim of success.

2.3.3 Research environment, team and resources

Host. The project is hosted by the **Institute of Global Health, Faculty of Medicine, University of Geneva**, with a methodological collaboration with **Data Science for Digital Health** (Prof. Douglas Teodoro) and a clinical one with **HUG emergency medicine** (Prof. Thibaut Desmettre and Dr Robert Larribau, CASU-144).

Institute of Global Health. Epidemiology, infectious-disease modelling and automated evidence extraction meet in one institute, which is where the gap became visible. Independence is organisational, as an independent programme alongside the Institute's groups (§2.6). **DS4DH** contributes the biomedical language processing that WP1 needs. Through GESICA I already work with the emergency-medicine partners, as the review of AI in emergency medical services [Edjinedja 2026] shows.

Existing infrastructure. The project builds on the LiteRev-Evidence prototype (§2.2.4), UNIGE HPC and the University's secure environment for clinical data.

Commitments, by status.

Item	Status	Evidence or action
Legionellosis linked data	**Secured**	BASEC 2026-00324, médecin cantonal and SIG data
Host institute (ISG)	**Agreed**	Hosting accepted, SNSF letter being signed
Computing	**Secured**	UNIGE HPC and secure environment in place
DS4DH methodological collaboration	**Secured**	Letter of support signed
SNSF institutional confirmation	Requested	UNIGE Vice-Rectorate, standard step
CASU-144 records, primary outcome	**Access held (GESICA)**	Daily, near real time, project agreement and letter in preparation
ED presentations and ICU occupancy	In progress	Extraction discussed with HUG, letters requested
COVID-19 hospitalisations (HUG, daily)	Re-request planned	Analysed before [Orel 2024], for validation and calibration
Wastewater and pharmacy sales (pharmacien cantonal)	Contacts established	Extracts to formalise, partly in GESICA
Open series (surveillance, wastewater, deaths, weather)	**Public**	No agreement needed

The remaining feasibility items are the ED and ICU extraction and the project agreement with 144. Every row rests on an existing relationship.

Resources requested. Project funds are capped at **CHF 250,000 over four years**, and the applicant's salary is covered separately. Doctoral students and postdocs cannot be employed under the 2026 rules, but support personnel can.

Item	Rationale
Scientific/technical collaborator, **50% over 48 months**	T1.2 protocol and adjudication support, WP3 harmonisation and evaluation pipeline, reproducibility
Second independent extractor	Contracted M3–M9, required for T1.2's dual-extraction design
Data access, secure storage and processing	Project-specific extractions and secure computing beyond existing infrastructure
Travel, conferences, incoming visits	Presentation of results, international collaboration
Other eligible direct costs	As justified in the final SNSF budget

Preparatory work, before month 1 and outside grant funding. It covers **project-specific data agreements** with HUG and 144/CASU, which re-scope access already held, and **project-specific data governance and reuse**, confirmed under the applicable institutional and regulatory framework before any analysis. It also covers the **hardening of LiteRev-Evidence** before clinical data connect. Feasibility rests on **the prototype that already exists** and on the fallbacks, each of which leaves the central question answerable.

2.4 Schedule and milestones

Who does what. I carry out WP2 and the confirmatory design of WP3 myself, and I lead WP1's protocol and WP4's decision work. The **collaborator** (50% over 48 months) carries the T3.1–T3.2 harmonisation, the replay pipeline and the reproducibility engineering behind D1.1 and D3.1, and the **independent extractor** (M3–M9) provides T1.2's second extraction. No other personnel are requested.

No task waits on a single chain. WP2 falls back to weakly informative priors if WP1 finds extraction inadequate, WP3's open-data track runs whatever the data access, and WP4's decision analysis is retrospective. **Open data alone has a cost.** Open series do not measure demand on emergency services, so the claim narrows and most of WP4 becomes illustrative. This is why the agreements start before the award.

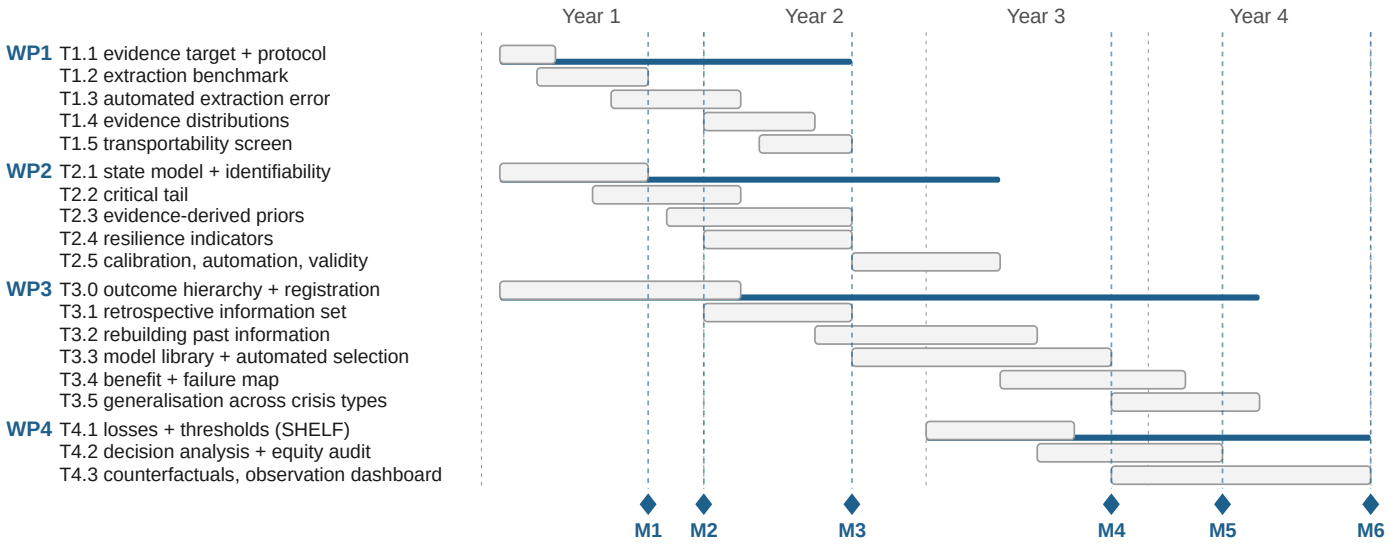


Figure 2. Work packages, tasks and milestones over 48 months.

Milestone	Month	Criterion
M1	9	Extraction benchmark released
M2	12	State model passes identifiability criteria, or fallback selected
M3	20	Open-data benchmark running, operational series harmonised, or fallback activated
M4	34	Primary cold-start hypothesis tested by the champion comparison
M5	40	Decision relevance established
M6	48	Analysis across crisis types complete, dashboard running and scored prospectively for validated demand-driven crisis types where authorised

2.5 Relevance and impact

A null result changes identifiable practice. Modellers seed early-crisis models with parameters taken from a few studies, and evidence-synthesis platforms (mine included) assume that pooled

estimates can be reused as priors. A calibrated null tells both that this practice gains nothing where it matters most.

Form of publication. Peer-reviewed articles, preprints on submission and open access throughout. Benchmarks, the prior library and the code are released as **citable, versioned open resources**, and results are returned to the partners. Durable resources, subordinate to the central question, are **the extraction and replication benchmark**, **the public cold-start benchmark**, an **open reference framework** and a **decision-analytic evaluation framework**.

Practical and societal impact. In the first weeks of a crisis, emergency services decide whether to open capacity, move resources or escalate before local outcomes are informative. The project measures whether external evidence improves those decisions, and with how much remaining uncertainty. Each validated crisis type leaves a configuration that re-validates itself and, where authorised, a running dashboard with the services. Extending this core to several types of crisis is the explicit path after the grant.

2.6 Relevance for personal career development

Why I am ready, and why now. I have the methods, the prototype and access to the domain, used so far inside other people's programmes. What I lacked is a research agenda of my own.

Programme	Objective	My role	Distinction from Ambizione
GESICA	Franco-Swiss crisis intelligence	Contributor	Decision support, no test of evidence priors
GeoAI4EI (Horizon Europe)	Epidemic-intelligence toolbox	Contributor	Infrastructure, no test of borrowing
Legionellosis study (ongoing)	Waterborne outbreaks in Geneva	Data lead	Supplies a contrasting extension
Ambizione	**When borrowed evidence helps at cold start, and when it harms**	**PI, full time**	-

This project does not build another epidemic-intelligence platform. It uses existing infrastructures to test a question that is a work package of none of them.

The question and the methods are mine. The question arose while I was building LiteRev-Evidence. The platform could pool published estimates into priors, and nothing showed whether they improved forecasting in a new setting. The methodological core came from fifteen years of quantitative finance, not from any group I have worked in, and the platform was developed at UNIGE under my scientific direction.

Collaborators and their roles. Prof. Olivia Keiser leads the group in which I developed LiteRev. She also leads the legionellosis study, GeoAI4EI and the MAS on which I teach, and she co-authors my output, as expected at this stage, but she has **no scientific role in this project**. Prof. Douglas Teodoro leads GESICA and provides the WP1 collaboration on biomedical language processing.

The organisational position is the transition. Hosted by the **Institute of Global Health**, I will lead an independent programme alongside its groups rather than inside one, and the confirmation letters set out the guarantees. The programme can be carried out by the PI and the budgeted support personnel. **No named individual other than the PI is scientifically load-bearing.**

Where this leads. By month 48 I will have a validated answer and a first-author record, the basis for leading a group or holding a professorship.

Bibliography

Surveillance, nowcasting and outbreak detection

- **[Farrington 1996]** Farrington CP, Andrews NJ, Beale AD, Catchpole MA. A statistical algorithm for the early detection of outbreaks of infectious disease. *Journal of the Royal Statistical Society Series A* 1996;159(3):547–563.
- **[Noufaily 2013]** Noufaily A, Enki DG, Farrington P, Garthwaite P, Andrews N, Charlett A. An improved algorithm for outbreak detection in multiple surveillance systems. *Statistics in Medicine* 2013;32(7):1206–1222. doi:10.1002/sim.5595
- **[Le Strat 1999]** Le Strat Y, Carrat F. Monitoring epidemiologic surveillance data using hidden Markov models. *Statistics in Medicine* 1999;18(24):3463–3478.
- **[Watkins 2009]** Watkins RE, Eagleson S, Veenendaal B, Wright G, Plant AJ. Disease surveillance using a hidden Markov model. *BMC Medical Informatics and Decision Making* 2009;9:39. doi:10.1186/1472-6947-9-39
- **[Höhle 2014]** Höhle M, an der Heiden M. Bayesian nowcasting during the STEC O104:H4 outbreak in Germany, 2011. *Biometrics* 2014;70(4):993–1002. doi:10.1111/biom.12194
- **[McGough 2020]** McGough SF, Johansson MA, Lipsitch M, Menzies NA. Nowcasting by Bayesian smoothing: a flexible, generalizable model for real-time epidemic tracking. *PLoS Computational Biology* 2020;16(4):e1007735. doi:10.1371/journal.pcbi.1007735
- **[Lazer 2014]** Lazer D, Kennedy R, King G, Vespignani A. The parable of Google Flu: traps in big data analysis. *Science* 2014;343(6176):1203–1205. doi:10.1126/science.1248506

Prehospital data as a leading indicator

- **[Rosenkötter 2013]** Rosenkötter N, Ziemann A, Garcia-Castrillo Riesgo L, Gillet JB, Vergeiner G, Krafft T, Brand H. Validity and timeliness of syndromic influenza surveillance during the autumn/winter wave of A(H1N1) influenza 2009: results of emergency medical dispatch, ambulance and emergency department data from three European regions. *BMC Public Health* 2013;13:905. doi:10.1186/1471-2458-13-905
- **[Bonora 2025]** Bonora R, Ticozzi EM, Pregliasco FE, Pagliosa A, Bodina A, Cereda D, Perotti G, Lombardo M, Stirparo G. Telephone calls to emergency medical service as a tool to predict influenza-like illness: a 10-year study. *Public Health* 2025;238:239–244. doi:10.1016/j.puhe.2024.12.021
- **[Edjinedja 2026]** Edjinedja KL, Larribau R, Orel E, Cossus J, Elfahim O, Yendouname K, Vaussenat F, Teodoro D, Robert-Nicoud S, Barakat O, Desmettre T. Artificial intelligence in emergency medical services for disasters and health emergencies: a systematic review. Submitted, 2026.

Collaborative forecasting and its evaluation

- **[Cramer 2022]** Cramer EY, Ray EL, Lopez VK, et al. (consortium, more than 50 authors). Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the United States. *Proceedings of the National Academy of Sciences* 2022;119(15):e2113561119. doi:10.1073/pnas.2113561119
- **[Sherratt 2023]** Sherratt K, Gruson H, Grah R, et al. (consortium, more than 50 authors). Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. *eLife* 2023;12:e81916. doi:10.7554/eLife.81916

Demand forecasting and operational surveillance tooling

- **[Wargon 2009]** Wargon M, Guidet B, Hoang TD, Hejblum G. A systematic review of models for forecasting the number of emergency department visits. *Emergency Medicine Journal* 2009;26(6):395–399. doi:10.1136/emj.2008.062380
- **[Salmon 2016]** Salmon M, Schumacher D, Höhle M. Monitoring count time series in R: aberration detection in public health surveillance. *Journal of Statistical Software* 2016;70(10):1–35. doi:10.18637/jss.v070.i10

Evidence as prior information

- **[Cook 2023]** Cook JD, Williams DM, Walsh DP, Hefley TJ. Bayesian forecasting of disease spread with little or no local data. *Scientific Reports* 2023;13:8137. doi:10.1038/s41598-023-35177-6
- **[Ibrahim 2000]** Ibrahim JG, Chen M-H. Power prior distributions for regression models. *Statistical Science* 2000;15(1):46–60.
- **[Hobbs 2011]** Hobbs BP, Carlin BP, Mandrekar SJ, Sargent DJ. Hierarchical commensurate and power prior models for adaptive incorporation of historical information in clinical trials. *Biometrics* 2011;67(3):1047–1056.
- **[Schmidli 2014]** Schmidli H, Gsteiger S, Roychoudhury S, O'Hagan A, Spiegelhalter D, Neuenschwander B. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics* 2014;70(4):1023–1032. doi:10.1111/biom.12242

Automated extraction and living reviews

- **[Elliott 2014]** Elliott JH, Turner T, Clavisi O, Thomas J, Higgins JPT, Mavergames C, Gruen RL. Living systematic reviews: an emerging opportunity to narrow the evidence-practice gap. *PLoS Medicine* 2014;11(2):e1001603. doi:10.1371/journal.pmed.1001603
- **[Shankar 2026]** Shankar R, Lim A, Qian X. Performance of large language models in data extraction for evidence synthesis: a systematic review. *Journal of Biomedical Informatics* 2026;181:105086. doi:10.1016/j.jbi.2026.105086
- **[Orel 2023]** Orel E, Ciglenecki I, Thiabaud A, Temerev A, Calmy A, Keiser O, Merzouki A. An automated literature review tool (LiteRev) for streamlining and accelerating research using natural language processing and machine learning. *Journal of Medical Internet Research* 2023;25:e39736. doi:10.2196/39736

Transportability

- **[Bareinboim 2016]** Bareinboim E, Pearl J. Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences* 2016;113(27):7345–7352. doi:10.1073/pnas.1510507113
- **[Dahabreh 2020]** Dahabreh IJ, Petito LC, Robertson SE, Hernán MA, Steingrimsson JA. Toward causally interpretable meta-analysis: transporting inferences from multiple randomized trials to a new target population. *Epidemiology* 2020;31(3):334–344. doi:10.1097/EDE.0000000000001177
- **[Degtiar 2023]** Degtiar I, Rose S. A review of generalizability and transportability. *Annual Review of Statistics and Its Application* 2023;10:501–524.

Critical slowing down and resilience indicators

- **[Scheffer 2009]** Scheffer M, Bascompte J, Brock WA, Brovkin V, Carpenter SR, Dakos V, Held H, van Nes EH, Rietkerk M, Sugihara G. Early-warning signals for critical transitions. *Nature* 2009;461:53–59. doi:10.1038/nature08227
- **[O'Regan 2013]** O'Regan SM, Drake JM. Theory of early warning signals of disease emergence and leading indicators of elimination. *Theoretical Ecology* 2013;6:333–357.
- **[Brett 2018]** Brett TS, O'Dea EB, Marty É, Miller PB, Park AW, Drake JM, Rohani P. Anticipating epidemic transitions with imperfect data. *PLoS Computational Biology* 2018;14(6):e1006204. doi:10.1371/journal.pcbi.1006204
- **[Southall 2021]** Southall E, Brett TS, Tildesley MJ, Dyson L. Early warning signals of infectious disease transitions: a review. *Journal of the Royal Society Interface* 2021;18(182):20210555. doi:10.1098/rsif.2021.0555

Conformal prediction

- **[Angelopoulos 2023]** Angelopoulos AN, Bates S. Conformal prediction: a gentle introduction. *Foundations and Trends in Machine Learning* 2023;16(4):494–591.
- **[Barber 2023]** Barber RF, Candès EJ, Ramdas A, Tibshirani RJ. Conformal prediction beyond exchangeability. *Annals of Statistics* 2023;51(2):816–845.

Model components: regimes, extremes and transmission

- **[Keeling 2008]** Keeling MJ, Rohani P. *Modeling Infectious Diseases in Humans and Animals.* Princeton University Press, 2008.
- **[Hamilton 1989]** Hamilton JD. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* 1989;57(2):357–384.
- **[Coles 2001]** Coles S. *An Introduction to Statistical Modeling of Extreme Values.* Springer, 2001. doi:10.1007/978-1-4471-3675-0
- **[Ranjbar 2022]** Ranjbar S, Cantoni E, Chavez-Demoulin V, Marra G, Radice R, Jaton-Ogay K. Modelling the extremes of seasonal viruses and hospital congestion: the example of flu in a Swiss hospital. *Journal of the Royal Statistical Society Series C (Applied Statistics)* 2022;71(4):884–905. doi:10.1111/rssc.12559

Heat, air quality and emergency demand

- **[Schulte 2024]** Schulte F, Röösli M, Ragettli MS. Risk, attributable fraction and attributable number of cause-specific heat-related emergency hospital admissions in Switzerland. *International Journal of Public Health* 2024;69:1607349. doi:10.3389/ijph.2024.1607349
- **[Ragettli 2019]** Ragettli MS, Vicedo-Cabrera AM, Schindler C, Röösli M. Impact of the warm summer 2015 on emergency hospital admissions in Switzerland. *Environmental Health* 2019;18:66. doi:10.1186/s12940-019-0507-1
- **[Zheng 2015]** Zheng XY, Ding H, Jiang LN, Chen SW, Zheng JP, Qiu M, Zhou YX, Chen Q, Guan WJ. Association between air pollutants and asthma emergency room visits and hospital admissions in time series studies: a systematic review and meta-analysis. *PLoS ONE* 2015;10(9):e0138146. doi:10.1371/journal.pone.0138146
- **[Lung 2021]** Lung S-CC, Yeh J-CJ, Hwang J-S. Selecting thresholds of heat-warning systems with substantial enhancement of essential population health outcomes for facilitating

implementation. *International Journal of Environmental Research and Public Health* 2021;18(18):9506. doi:10.3390/ijerph18189506

Evaluation and decision value

- **[Gneiting 2007]** Gneiting T, Raftery AE. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 2007;102(477):359–378.
- **[Vickers 2006]** Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making* 2006;26(6):565–574.

Own work cited in the plan

- **[Orel 2022]** Orel E, Esra R, Estill J, Thiabaud A, Marchand-Maillet S, Merzouki A, Keiser O. Prediction of HIV status based on socio-behavioural characteristics in East and Southern Africa. *PLoS ONE* 2022;17(3):e0264429. doi:10.1371/journal.pone.0264429
- **[Orel 2024]** Orel E, Sobel J, Zanella MC, Iten A, Kaiser L, Keiser O, Vetter P. Clinical severity of Delta and 3 different Omicron sublineages in patients hospitalized because of COVID-19 in a Swiss tertiary center. *CMI Communications* 2024;1(2):100005. doi:10.1016/j.cmicom.2024.100005
- **[Estill 2020]** Estill J, Venkova-Marchevska P, Roelens M, Orel E, Temerev A, Flahault A, Keiser O. Future scenarios for the SARS-CoV-2 epidemic in Switzerland: an age-structured model. *F1000Research* 2020;9.
- **[Merzouki 2021]** Merzouki A, Estill J, Orel E, Tal K, Keiser O. Clusters of sub-Saharan African countries based on sociobehavioural characteristics and associated HIV incidence. *PeerJ* 2021;9:e10660. doi:10.7717/peerj.10660
- **[Esra 2023]** Esra R, Carstens J, Le Roux S, Mabuto T, Eisenstein M, Keiser O, Orel E, Merzouki A, De Voux L, Maskew M, Sharpey-Schafer K. Validation and improvement of a machine learning model to predict interruptions in antiretroviral treatment in South Africa. *Journal of Acquired Immune Deficiency Syndromes* 2023;92(1):42–49.
- **[Nwosu 2021]** Nwosu KD, Fokam J, Wanda F, Mama L, Orel E, Ray N, Meke J, Tassegning A, Takou D, Mimbe E, Stoll B, Guillebert J, Comte E, Keiser O, Ciaffi L. SARS-CoV-2 antibody seroprevalence and associated risk factors in an urban district in Cameroon. *Nature Communications* 2021;12:5851. doi:10.1038/s41467-021-25946-0
- **[Ng'ambi 2026]** Ng'ambi W, Estill J, Keiser O, Merzouki FA, Orel E. Machine learning-based classification of self-reported cardiovascular disease history in Africa using harmonised multi-country WHO STEPS surveys: 2014–2019. Accepted for publication.